

Unitree Week 20

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This Week

- Goal is to complete the AMASS human motion dataset to G1 dataset code.

Additions to the Inverse Kinematics Code

- `Prepare_g1_target.py`: Goal: to prevent the robot from turning one way while arms/feet targets point another way.
 - `root_pos` = where the body moves
 - `root_yaw` = where the body faces
 - Local keypoints = body pose relative to the root
 - Also, saved more motion signals (left/right foot height, pelvis vertical speed, hand speeds, elbow speeds, foot speeds, yaw speed)
→ retarget code adapts weights for specific movements.
- `Retarget_g1.py`:
 - Loading the new keypoints from the `prepare_g1_target.py` code.
- Overall: More error diagnostics to find out what joints need tuning.

Current Issues with code

- With new tuning/adjustments, I was able to get the G1 to do the turning motion.
- However, it had worsened the jumping motion. → in the AMASS dataset, the motion follows a human turning and then jumping right after at the direction turned.
 - The G1 does not properly jump, the upper body of the G1 does not jump facing the proper direction.

06/12/2026 Updates

- Code was improved so that inverse kinematics works for two actions at once (turning + jumping action).
- Although there is still some minor problems occurring with specific joints, causing imperfect results.

Unitree G1 Retargeted Motion Datasets

- Provides motion data in a unified format suitable for training humanoid robot control policies.
 - Pipeline: Human Motion Capture in SMPL/SMPL-X format → Mink Inverse Kinematics Solver → Physics-Based Foot Contact Detection → Manual Inspection & Filtering
- **Can utilize these open-source datasets to do Imitation Learning**

References

<https://www.youtube.com/watch?v=g9dBdkwtURE>

<https://amass.is.tue.mpg.de/index.html>

<https://github.com/ZhengyiLuo/SMPLSim>

<https://mano.is.tue.mpg.de/index.html>

<https://huggingface.co/datasets/openhe/g1-retargeted-motions>